HySA (Hybrid Semantic Architecture): Preserving Polysemantic Abstraction via Contrastive Inference Tensors, Monosemantic Indexing, and Topological Temperature (T_t)

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October 2026

"The AI industry views polysemanticity as a curse... yet perhaps it is the very blessing that made intelligence possible."

Abstract

Contemporary *Mechanistic Interpretability* research predominantly treats neuronal polysemanticity as an undesirable artifact of dimensional compression (*the Superposition Hypothesis*), focusing on decomposing latent spaces into vast, sparse dictionaries of purely monosemantic features. In this work, we propose a paradigm shift: **polysemanticity is not a defect to be eradicated, but the geometric prerequisite for intelligence and analogical reasoning.** A purely monosemantic neuron is intrinsically "stupid" (zero intelligence): lacking cross-domain associative connections, it functions as a rigorous yet sterile memory cell. To decouple fluid associative behavior from deterministic factual verification without resorting to serial architectures (which introduce non-abstractive bottlenecks) or parallel networks (which are computationally inefficient), we introduce **HySA (*Hybrid Semantic Architecture*)**, a unified paradigm delimited by neuronal semanticity. HySA introduces four core contributions: (1) **Contrastive Inference Superposition** (I_n), extracting the multi-concept cube of polysemantic units via weighted inference expectations over contrastive datasets; (2) **Monosemantic Indexing**, coupling a polysemantic network (N_p, intelligence) with an indexed registry of monosemantic centroids (N_m/N_k, memory) queried via deterministic internal APIs; (3) the **Semantic Hypersphere**, governed by dual temperatures (T_p, T_m) and **Topological Temperature** (T_t); and (4) the **Compression Gradient** (Δq) between N_p and N_m, governed by the Degree of Determinism $\beta = T/q$ around a stationary **Critical Point** (P_c).

1 Introduction: Intelligence vs. Memory

In deep neural networks, individual units consistently exhibit *mixed selectivity* [1, 2]. Prevailing approaches utilizing *Sparse Autoencoders* (SAEs) [3, 4] seek to disentangle these representations into orthogonal, monosemantic dimensions. However, this pursuit of pure monosemanticity overlooks a fundamental epistemological division between **Intelligence** and **Memory**:

- **Polysemantic Neurons** (N_p) **Generate Intelligence:** A unit co-activating across concepts such as `IP`, `DNS`, `White Pages`, and `Phone Number` is not experiencing noisy interference. It is compressing multiple concepts into a unified higher-order meaning: the relational schema of *identity-to-address resolution*. **Associative connections are what make a system intelligent.**
- **Monosemantic Neurons** (N_m) **Provide Memory:** Units encoding rigid atomic literals—such as the IP address `8.8.8.8`, the integer `24`, or the proper noun `JOHN`—possess zero associative connections ($\varnothing$ intelligence). Like entries in a static dictionary, a monosemantic unit is cognitively "stupid," yet it is **strictly rigorous**—making it essential for preventing hallucinations.

Modern LLMs hallucinate because they entangle both regimes within homogeneous weight matrices [5], allowing associative gradients to interpolate across immutable facts.

2 Clarification of Systems Terminology

Each indexed unit N_k possesses a monosemantic centroid μ_k structurally analogous to an operating system **"Inode"**, which motivates our use of the term **"Filesystem"**. To prevent cross-disciplinary ambiguity, Table 1 maps these systems heuristics to their formal mathematical constructs in latent space $\mathcal{S} \subset \mathbb{R}^D$.

3 Contrastive Superposition Tensors (I_n)

A polysemantic neuron functions as a high-dimensional superposition cube compressing multiple concepts into unified meanings. Because an isolated neuron is a mute scalar, its internal geometry is extracted as the **weighted average of all inference outputs across a contrastive dataset** ($\mathcal{X}^+$ containing shared structural in-

Table 1: Mapping Systems Heuristics to Formal Constructs.

Systems Term	Formal Construct	Architectural Role
"Inode" (N_k)	Centroid $\mu_k \in \mathbb{R}^D$	Orthogonal pointer to an immutable atomic literal (e.g., `8.8.8.8`).
"Filesystem"	Indexed Manifold $\mathcal{F}_{\text{mono}} = \{\mu_1, \dots, \mu_M\}$	Dictionary storage ($\varnothing$ intelligence, 100% rigor) for verification.
"Query / API"	Metric Operator $Q(q, \mathcal{F}_{\text{mono}}, T_t)$	Deterministic boundary check confirming polysemantic states.

variants vs. $\mathcal{X}^-$ baseline):

$$I_n = \frac{1}{\sum_k z_{k,n} + \eta} \Bigg[\sum_{X_k \in \mathcal{X}^+} z_{k,n} \cdot \Phi_{\text{net}}(X_k \mid n) - \gamma \sum_{X_j \in \mathcal{X}^-} z_{j,n} \cdot \Phi_{\text{net}}(X_j \mid n) \Bigg] \tag{1}$$

where $z_{k,n}$ is unit n's activation and $\Phi_{\text{net}}(\cdot \mid n)$ is the whole-network inference conditioned on n. Shared invariants are queried via $I_A \odot I_B$ and divergences via $\text{ReLU}(I_A - I_B)$ (Figure 1).

4 The Unified Hybrid Paradigm (HySA)

How should polysemantic intelligence (N_p) and monosemantic memory (N_m) interact?

- **Neither Serial:** Placing a rigid monosemantic layer downstream of N_p creates a *non-abstractive bottleneck* that destroys contextual relations.
- **Nor Parallel:** Executing duplicate networks concurrently is *extremely inefficient* computationally.

In **HySA**, the model is a **unified whole**, delimited internally by logic and by the specific "semanticity" (dimensionality and cardinality M) of individual neurons within the layers. Fluid reasoning propagates through N_p; when factual confirmation is required, the trajectory executes a concise internal **Query/API** to the indexed monosemantic centroids $\mathcal{F}_{\text{mono}}$, imparting deterministic consistency.

5 The Semantic Hypersphere and Dual Temperatures (T_p, T_m, T_t)

Rather than applying temperature as post-hoc *Softmax* noise, HySA defines temperature geometrically as **the multidimensional neighborhood of the neuron**: a **Semantic Hypersphere** in latent space.

When a latent trajectory is *queried* within this hypersphere (where related tokens or concepts such as `CAT`, `GATTO`, `KG`, or `MACCHINA` lie along radial vectors A, B, C, D), system dynamics are governed by two regime temperatures—the **Polysemantic Temperature** (T_p) and **Monosemantic Temperature** (T_m)—bounded by the **Topological Temperature** (T_t):

$$B(\mu_k, T_t) = \{x \in \mathcal{S} \mid \|x - \mu_k\|_2 \leq T_t(T_p, T_m)\} \tag{2}$$

If a queried point q falls within radius T_t of centroid μ_k, the API confirms the reasoning step and binds the exact literal from μ_k. If $\|q - \mu_k\|_2 > T_t$, HySA raises a deterministic *Semantic Fault*, halting hallucination (Figure 2).

6 Compression Gradient (Δq), Determinism (β), and Critical Point (P_c)

The structural ratio between polysemantic neurons (N_p) and monosemantic centroids (N_m) defines the **Compression Gradient** (Δq) across layers. Let $q = f(N_p, N_m, M)$ represent the effective neuronal quantity. We define the **Degree of Determinism** ($\boldsymbol{\beta}$) as:

$$\beta = \frac{T}{q} \tag{3}$$

Mapping $\boldsymbol{\beta}$ against the compression gradient Δq yields a convex potential landscape with a stationary minimum at the **Critical Point** (P_c, Figure 2b):

1. **Left of P_c (Under-Constrained $\boldsymbol{\Delta q}$):** Dominance of N_p without sufficient N_m indexing leads to ungrounded associative drift.
2. **Right of P_c (Over-Indexed Saturation):** Dominance of N_m severs associative links; $\mathcal{V}_m \propto M \cdot (T_t)^D$ saturates the space, freezing the model into a static dictionary.
3. **At the Critical Point ($\boldsymbol{P_c}$):** Optimal equilibrium between N_p and N_m, maximizing polysemantic abstraction while guaranteeing deterministic verification.

7 Experimental Validation

We validate HySA across two complementary benchmarks:

1. 32×32 **Contrastive Benchmark:** A 4-neuron bottleneck trained on dual-nature binary matrices (shared digit loops + rigid 2×2 top-left literal checksums). As shown in Figure 1 and Figure 2c, algebraic queries ($I_1 \odot I_2$ and $\text{ReLU}(I_1 - I_2)$) isolate structural invariants, while the Query API maintains 0.0% **factual literal corruption** across $T_p \in [0.05, 2.5]$.
2. **Real-World Transformer Validation (`Pythia-70M`):** Probing Layer 3 residual streams ($D = 512$) on `EleutherAI/pythia-70m` confirms that isomorphic concepts (`DNS`, `IP Routing`, `White Pages`, `Phone Directory`) converge onto a shared polysemantic subspace with cosine similarities between **0.893** and **0.973** (Table 2). Indexing normalized monosemantic centroids (N_k) on S^{511} with radius $T_t = 0.48$ successfully confirms valid protocol queries ($d \in [0.324, 0.430] \leq T_t$) while

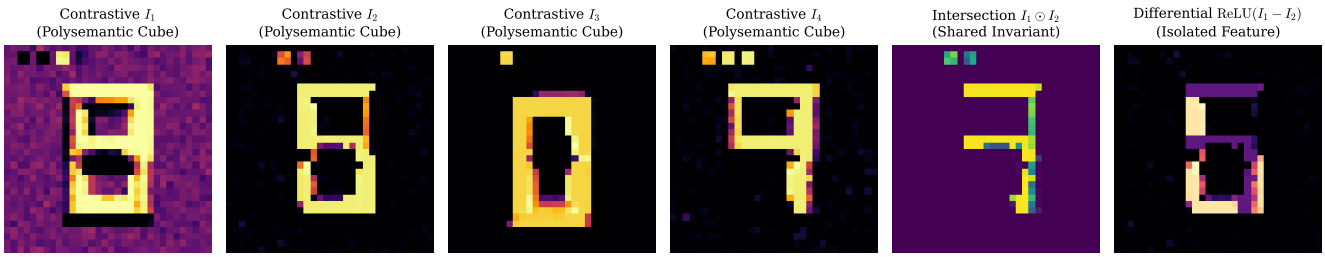


Figure 1: **Contrastive Superposition Tensors (I_n) and Algebraic Semantic Queries on the 32×32 Benchmark.** Columns 1–4 show polysemantic representations learned by a 4-neuron bottleneck. Column 5 isolates the shared invariant ($I_1 \odot I_2$). Column 6 extracts differential feature divergence ($\mathrm{ReLU}(I_1 - I_2)$).

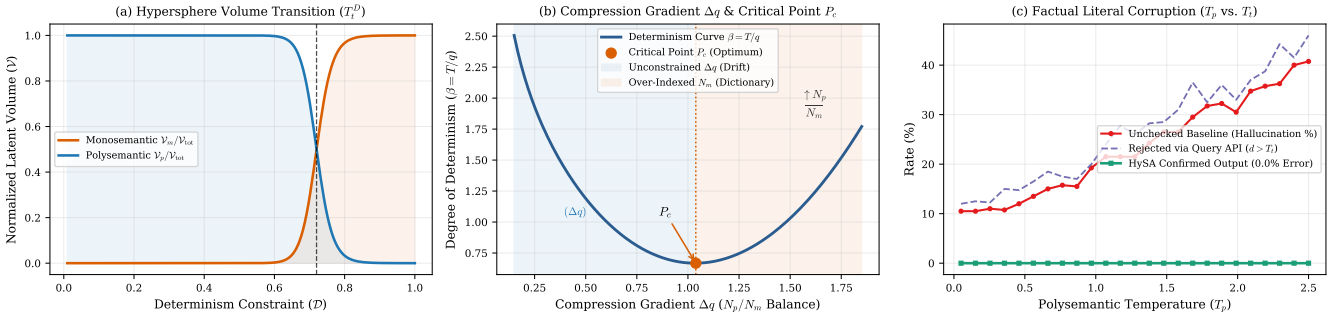


Figure 2: **High-Dimensional Volume Phase Transition, Critical Point (P_c), and Factual Verification in HySA. (a)** Step-function transition between Free Polysemantic Volume $\mathcal{V}_p$ and Monosemantic Constraint Volume $\mathcal{V}_m$. **(b)** Compression gradient Δq potential curve displaying the stationary Critical Point P_c over $\beta = T/q$. **(c)** Unchecked decoding suffers climbing factual corruption, whereas HySA maintains 0.0% corruption via monosemantic API confirmation within T_t.

intercepting unverified subterranean telemetry queries via a deterministic *Semantic Fault* ($d = 0.497 > T_t$).

Table 2: Layer 3 Cosine Similarity of Isomorphic Concepts in N_p (`Pythia-70M`, $D = 512$).

Concept in N_p	**DNS**	**IP Routing**	**White Pages**	**Phone Dir.**
DNS	1.000	0.961	0.973	0.893
IP Routing	0.961	1.000	0.940	0.916
White Pages	0.973	0.940	1.000	0.903
Phone Dir.	0.893	0.916	0.903	1.000

8 Conclusion

HySA demonstrates that polysemanticity is the core engine of neural intelligence (N_p), while monosemanticity is the rigorous guardian of factual memory (N_m). By coupling both regimes within a unified architecture governed by the Semantic Hypersphere (T_t) at the Critical Point (P_c), models achieve creative generalization with deterministic factual consistency.

References

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